Solutions to Exploration P.3

1. 1/3
2. Answers will vary. …. "Chance you pick the correct door."
3. Answers will vary. … Example follows:

Game #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Outcome (car or goat)	C	G	G	G	G	C	G	G	C	G	C	G	G	G	G

4. Answers will vary. 4/15 (from Example In 3.) Yes, this is close to 1/3, which is what I would expect just by random chance.
5. 0.333 (1/3). In the long run, this is the proportion of times I would expect to win by random chance.
6. Answers will vary. Example follows:

Number of games	10	20	30	40	50	60	70	80	90	100
Proportion of wins	0.40	0.35	0.26	0.25	0.34	0.31	0.32	0.35	0.33	0.31

7. The proportion of wins seems to jump around more for smaller numbers of games played, but as the number of games played increases, the proportion of wins settles. (in this example, around 0.45)
8. Answers will vary. Example follows. Yes, this is close to the 0.333 from #5.

Number of games	**100**	**200**	**300**	**400**	**500**	**600**	**700**	**800**	**900**	**1000**
Proportion of wins	0.31	0.31	0.32	0.33	0.34	0.34	0.33	0.34	0.34	0.34

9. Answers will vary. Some will say probabilities are both 0.333; others will think they may be different.
10. Answers will vary. Example follows:

Repetition	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Outcome (car or goat)	G	G	G	C	C	C	C	G	C	G	G	C	C	G	G

11. 7/15. It is higher than the proportion of 4/15 when stayed with original door instead of switching.

12. Answers will vary. Example follows:

	"Stay" strategy	"Switch" strategy
Wins (cars)	319	692
Losses (goats)	681	308
Total	1,000	1,000

13. One thousand is a pretty large number of repetitions and it appears that the switch strategy is superior. The proportion of times the car is won with the switch strategy is higher than the proportion of times the car is won with the stay strategy.
14. Estimate 0.692 for probability of winning with switch strategy.
15. Do more than 1,000 repetitions.
16. If you played the game forever, you would expect to win 2/3 of the games played. In other words, the long-run proportion of times you would win is 2/3.
17. Choose Door #3 initially, then always switch. You will only lose when Door #3 has the car which happens 10% of the time. So your probability of winning with this strategy is 0.90.